

Multi-Modal EEG Feature Fusion for Upper-Limb Motor State Classification: A Comparative Deep Learning Study

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Abstract

Single-trial classification of graded upper-limb movements from scalp EEG remains difficult because of low signal-to-noise ratio, spatial smearing due to volume conduction, and high inter-trial variability. This work systematically asks whether engineered spectral summaries and pairwise oscillatory coupling—in addition to a strong temporal convolutional baseline—raise multiclass accuracy on publicly available movement-execution EEG [1].

We compare a Common Spatial Patterns (CSP) plus Linear Discriminant Analysis (LDA) pipeline, an EEGNet-style convolutional model [2], and two dual-branch networks that fuse raw waveforms with either Welch power spectral density (PSD) maps or magnitude features from wavelet complex cross-spectral density at motor-relevant frequencies (10 and 22 Hz in our implementation).

Training followed the repository preprocessing and optimisation scripts ($\approx 85/15$ stratified split, subjects pooled).

Held-out accuracy increased from CSP ($\sim 38\%$), to EEGNet ($\sim 56\%$), to PSD fusion ($\sim 61\%$), and to connectivity attention fusion ($\sim 61\text{--}62\%$), so multimodal fusion adds only a modest increment beyond the temporal branch. Connectivity topographies contrasting hand-related movement epochs with rest nevertheless retain structured spatial contrast—evidence that ancillary features carry neurophysiologically interpretable information despite the limited classification gains. Finally, two-dimensional t-SNE embeddings of penultimate activations show persistent class overlap across architectures, aligning with an optimisation narrative in which variance-rich temporal pathways absorb most of the learning signal.

Index Terms—EEG; motor decoding; deep learning; multimodal fusion; functional connectivity; common spatial patterns.

1 Introduction

Decoding kinematic distinctions within a single limb from scalp electroencephalography (EEG) is relevant for assistive robotics, prosthetic control, and neurorehabilitation brain–computer interfaces (BCIs) [1]. At the scalp, cortical potentials are mixtures of underlying sources diluted by volume conduction; moreover, cue-locked trials exhibit low signal-to-noise ratio and stochastic latency jitter [1].

Classical pipelines employ spatial filtering—notably Common Spatial Patterns (CSP) combined with discriminant analysis—or band-power features tailored to μ and β rhythms. Convolutional EEG models such as EEGNet [2] jointly learn spectro-temporal patterns from waveform tensors and routinely outperform shallow filters on multiclass setups with sufficient data.

Objective. Rather than benchmarking for maximum leaderboard accuracy alone, we *systematically evaluate whether explicit Welch PSD summaries and pairwise wavelet connectivity features*

improve four-class upper-limb decoding beyond an EEGNet-style temporal encoder. The analysis is grounded in the movement-execution corpus of Ofner *et al.* [1], which provides controlled sustained upper-limb tasks at high temporal resolution together with ancillary documentation for event codes and sensor layouts.

Contributions. The study (i) implements a homogeneous preprocessing and splitting protocol spanning CSP+LDA [3], EEGNet, late PSD fusion (EEGPsdNet), and attention-based connectivity fusion (EEGConnectivityNet); (ii) quantifies where accuracy gains saturate once a temporal convolutional pathway is competitive; (iii) complements quantitative scores with scalp maps that contrast connectivity-derived summaries between *hand movement* epochs and *rest*, underscoring that engineered features preserve interpretable spatial structure even when classification margins are narrow; and (iv) visualises latent geometry via t-SNE on pre-softmax embeddings to explain why multimodal margins remain tight.

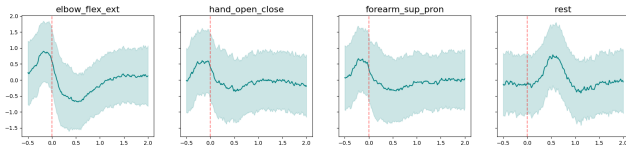


Figure 1: Grand-average cue-locked waveforms for the four coarse classes after the training pipeline’s filtering and normalisation. Similar sensorimotor morphologies across conditions indicate that single-trial discrimination must rely on subtle differences rather than separated templates.

Figure 1 illustrates cue-locked grand averages that foreshadow multiclass overlap.

2 Materials and methods

2.1 Dataset

We analysed publicly released *movement execution* EEG from [1] (BNCI Horizon 2020 accession 001-2017; see also docs/DatabaseDescription.pdf). Healthy adults performed cue-guided sustained actions of elbow, forearm, and hand synergy families together with fixation *rest*; recordings sampled 61 EEG channels at 512 Hz with auxiliary kinematics [1].

Each run corresponds to an EEGLAB ME_Sxx_rYY.mat archive read by EEGMatLoader. We retain 21 motor ribbon electrodes defined in src/config.py, resample to 128 Hz, and extract epochs from -0.5 s to $+2.0$ s relative to cue onset. Trials exceeding $\pm 200 \mu\text{V}$ amplitude limits are discarded.

Aggregation to four multiclass targets. Opposite movement directions collapse to symmetric families: elbow flexion/extension $\rightarrow$ *elbow*; forearm supination/pronation $\rightarrow$ *forearm*; hand open/close $\rightarrow$ *hand*; *rest* remains singleton (event codes enumerated in EVENT_ID). Chance performance is therefore 25%.

2.2 Preprocessing and derived tensors

The export script scripts/build_dataset.py applies EEGPreprocessor with notch filtering at line frequency, followed by zero-phase FIR band-pass filtering between 0.5 and 40 Hz (MNE; phase=’zero’). **Note:** exported tensors omit common-average re-referencing (apply_car=False) to match the convolutional branch; CSP hyperparameter sweeps in scripts/tune_csp_preprocessing.py may

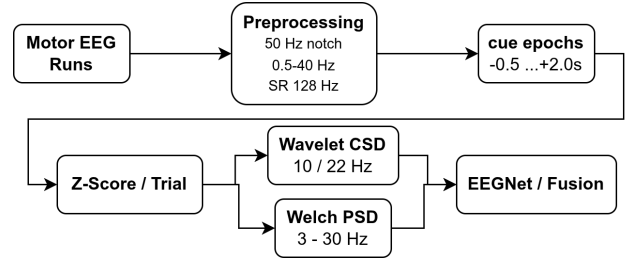


Figure 2: End-to-end feature flow from continuous recordings through optional spectral and connectivity branches into convolutional classifiers.

employ alternative band definitions and should be interpreted as exploratory rather than strictly identical preprocessing.

For each surviving trial:

- raw tensors undergo per-channel, per-trial temporal z-scoring prior to PSD and coherence extraction;
- **PSD tensors** (X_{psd}): Welch estimates over 3–30 Hz;
- **connectivity tensors** (X_{conn}): log-transformed magnitude of complex Morlet cross-spectral density at 10 and 22 Hz for upper-triangular channel pairs (src/processing.py), followed by dataset-wide standardisation.

Figure 2 summarises integration of optional branches into convolutional fusion networks. Stratified splitting (train $\approx 85\%$, test $\approx 15\%$, random seed 42) preserves class proportions.

2.3 Models and optimisation

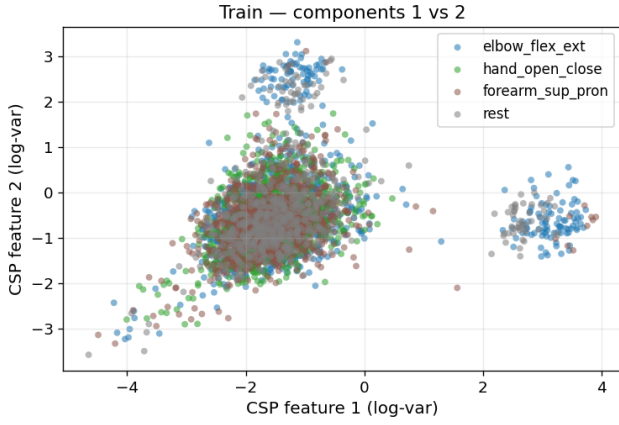
CSP+LDA. Six spatial filters with logarithmic energies feed scikit-learn Linear Discriminant Analysis within an MNE/decoding CSP pipeline [3].

Convolutional baselines. EEGNet adopts separable convolutional motifs from [2] as implemented in src/networks.py.

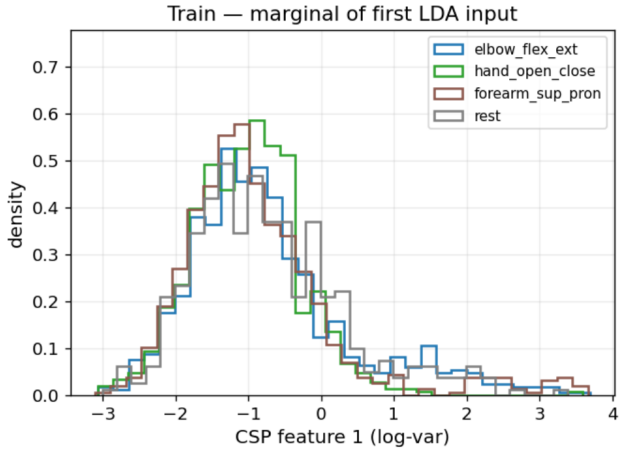
EEGPsdNet couples the EEGNet waveform encoder with a lightweight spectral CNN on X_{psd} ; representations are fused by vector concatenation and batch normalization before the terminal linear layer.

EEGConnectivityNet compresses waveform and connectivity embeddings, applies a softmax attention gate to mix modalities, and passes the fused vector through a multilayer perceptron classifier.

Optimisation. All deep models minimise class-weighted cross-entropy with label smoothing, op-



(a) Scattergram of CSP log-energies.



(b) Marginal density of an exemplar CSP feature.

Figure 3: Overlap of handcrafted CSP-derived features illustrates why linear separators alone achieve limited multiclass discrimination.

timise with AdamW (stepped learning-rate reduction on plateau), inject Gaussian augmentation noise on EEG tensors during training, and clip gradient norms; see `scripts/train_eeg*.py`.

2.4 t-SNE of penultimate embeddings

Activation vectors feeding the terminal affine layer (pre-softmax) were extracted for withheld trials across architectures. We projected them using scikit-learn’s TSNE with matched hyper-parameters to obtain comparable two-dimensional neighbourhoods.

2.5 Connectivity scalp visualisation

For qualitative neurophysiological corroboration, trial-averaged connectivity-derived quantities were interpolated to scalp coordinates to visualise contrasts between *hand* movement epochs and *rest*, using the Graz channel montage [1].

Table 1: Held-out multiclass accuracy (four coarse motor states vs. rest); chance = 25%.

Architecture	Representations	Accuracy [%]
CSP+LDA [3]	CSP log-energy (six filters)	~ 38 [†]
EEGNet [2]	raw waveform convolution	~ 56
EEGPsdNet	waveform + Welch PSD (late fuse)	~ 61
EEGConnNet [‡]	waveform + Morlet CSD (attention)	~ 61–62

[†]Higher CSP (38.5%) appears under narrow-band sweeps `tune_csp_grid.json`; CSP on exported wide-band tensors can fall nearer 32% (`csp_lda.ipynb`).

[‡]Implementation EEGConnectivityNet in `src/networks.py`.

3 Experimental results

3.1 Evaluation protocol and classification accuracy

Peak validation checkpoints were monitored with `scripts/train_*.py`. Unless stated otherwise, we report stratified pooling of ~85% training and ~15% withheld validation segments with identical random seed across models. All subjects contribute to both partitions; therefore the protocol indexes pooled mixing across participants rather than the leave-one-subject-out evaluation advocated for clinical extrapolation [1].

Table 1 summarises withheld accuracy ranks. CNN temporal encoding improves markedly relative to CSP ($\approx +18$ percentage points), whereas PSD concatenation contributes roughly five additional points ($\approx 56 \rightarrow 61\%$). Connectivity attention aligns with PSD fusion ($\approx 61\text{--}62\%$), indicating that multimodal augmentation alone does not reposition the optimisation landscape enough to transcend the EEGNet plateau—consistent with the hypothesis that temporal receptive fields disproportionately dominate learning dynamics.

3.2 Connectivity topographies contrasting hand movement and rest

Connectivity-derived contrasts between *hand* synergy epochs and *rest* reveal structured spatial modulation over motor-centred electrodes [1]: sensorimotor band coupling differentiates behavioural states despite limited downstream classifier separation. Hence neurophysiological plausibility and statistical classification performance furnish complementary diagnoses—motor-network fingerprints can remain visible while discriminative variance for fine-grained multiclass boundaries lingers predominantly in nuisance-rich temporal residuals.

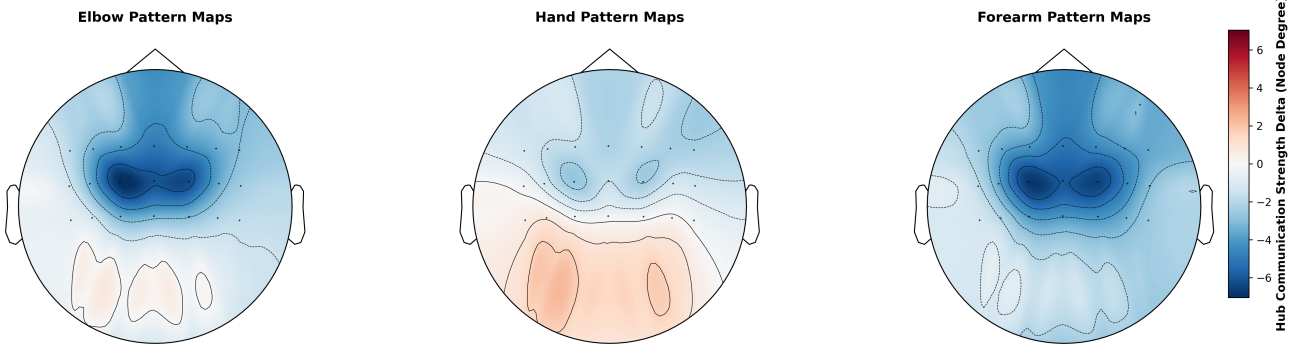


Figure 4: Connectivity-based scalp maps contrasting hand-related movement epochs with resting baselines (trial-averaged coupling summaries mapped to electrodes; Graz montage [1]). Contralateral and peri-central modulation indicates that coherence-inspired features encode task-relevant structure even though end-to-end accuracies saturate near the EEGNet PSD baseline.

3.3 Latent geometry before softmax

Penultimate embeddings were nonlinearly embedded with identical t-SNE hyper-parameters (Figure 5). Colours remain interleaved across EEGNet and both fusion architectures, confirming that softmax inputs occupy partially overlapping neighbourhoods. Interpretation remains qualitative because t-SNE nonlinearly warps distances; nevertheless pervasive colour mixing parallels the accuracy plateau induced by multimodal fusion.

3.4 Confusion structure

Diagonal concentration improves from CSP through CNN fusion, yet off-diagonal leakage persists predominantly among neighbouring synergies (Figure 6), echoing Figures 4–5. Training-validation divergence in archived learning curves `models/*/learning_curves.png` indicates that waveform branches reduce training losses faster while auxiliary branches saturate earlier.

4 Discussion

Neurophysiological corroboration versus quantitative scores. Figure 4 presents structured spatial contrasts between aggregate *hand* movement epochs and resting baselines [1], reinforcing that connectivity-derived summaries capture task-relevant physiology. Table 1 nevertheless confirms that PSD and coherence fusion elevate accuracy only incrementally once a competitive temporal CNN is in place. Interpreted jointly, these observations support the view that scalar accuracy and neurophysiological readability are not interchangeable: movement–rest distinctions can re-

main prominent in feature space while synergy-level boundaries continue to challenge classifiers operating on scalp recordings.

Temporal-branch dominance. Smoothed PSD maps and log-scaled coherence tensors exhibit lower trial-to-trial variance than jitter-augmented waveforms. Mini-batch gradients therefore emphasise depthwise temporal receptive fields that specialise in peri-central voltage fluctuations, even when batch normalisation or softmax attention gates nominally reweight auxiliary branches.

Latent entanglement. Figure 5 shows interleaved class neighbourhoods immediately prior to softmax, providing a geometrical explanation for the accuracy plateau: widening fusion layers alone does not orthogonalise class manifolds absent additional data diversity or regularisation targeted at coherence representations.

Limitations. (i) **Preprocessing alignment:** exploratory CSP sweeps `tune_csp_*` exploit band hypotheses that differ from the CNN export path; transparent harmonisation is prerequisite for strict CSP–CNN comparisons. (ii) **Evaluation scope:** withholding trials from a mixed-subject pool overestimates reproducibility relative to leave-one-subject-out protocols recommended for assistive translation [1].

Future directions. Modality-regularised objectives, coherence-only diagnostic models, hierarchical decoders that isolate joint identity before fine movements, subject-specific adaptation, and augmentation policies operating on embedding manifolds merit investigation.

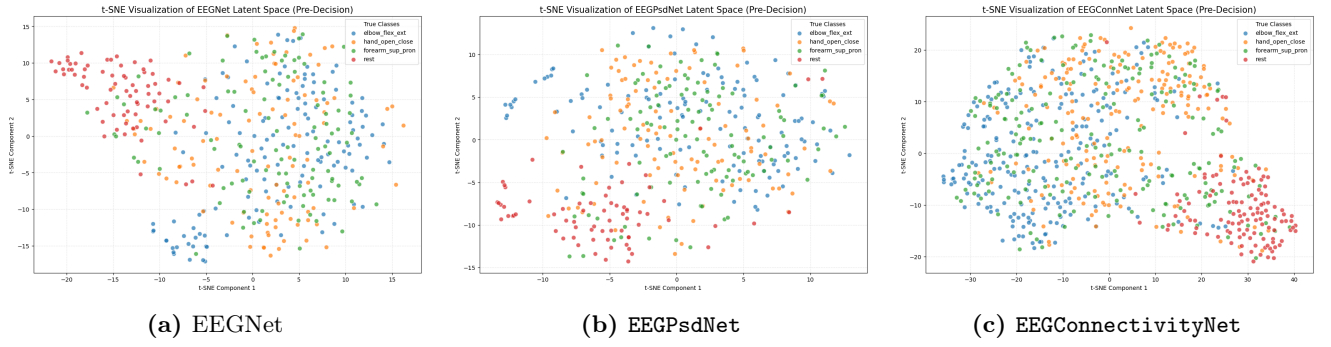


Figure 5: t-SNE visualisations of pre-softmax embeddings for withheld trials.

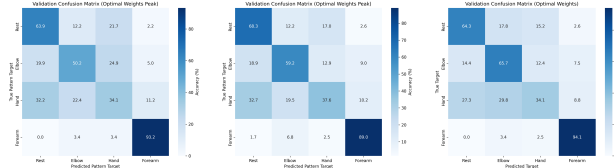


Figure 6: Row-normalised confusion matrices at peak validation checkpoints (percent allocated to each predicted label given a ground-truth row).

5 Conclusion

We conducted a systematic evaluation of whether PSD and wavelet connectivity features improve four-class upper-limb EEG decoding beyond an EEGNet-class temporal encoder [1, 2].

Deep convolutional filters deliver the principal accuracy gain relative to CSP+LDA [3]; PSD and attention-based connectivity fusion augment performance only modestly ($\approx 56\%$ to $\approx 61\text{--}62\%$).

Topographic contrasts between aggregated hand movements and resting baselines (Figure 4) retain organised peri-central modulation that is only partially conveyed by softmax scores, illustrating that multimodal augmentation may preserve interpretable physiology even when discriminative increments taper off.

Class-interleaved embeddings prior to softmax (Figure 5) provide a geometrical analogue of the same plateau under pooled-subject validation.

Future benchmarking should unify CSP and CNN preprocessing, adopt subject-independent validation, and explore regularisers that incentivise coherence pathways without weakening temporal stability.

Acknowledgment

Scripts, notebooks, raster figures (`report/img/`), and training artefacts (`models/`, `results/`) accompany the codebase.

References

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